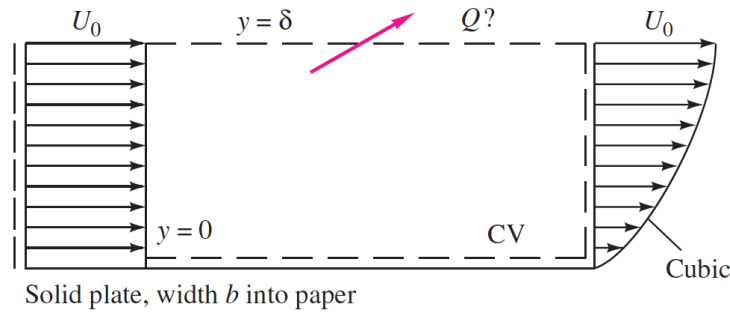
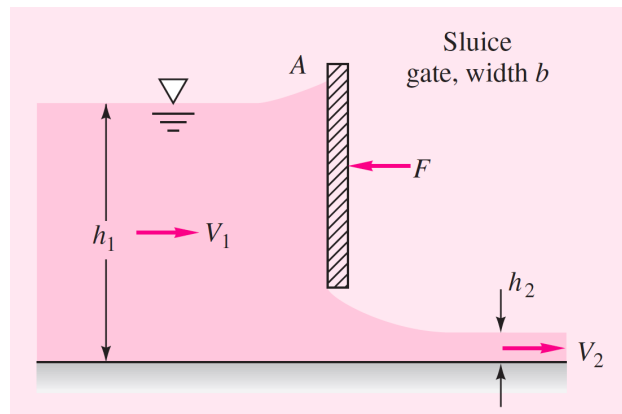


Problem 1: Incompressible fluid flows past an impermeable flat plate, with a uniform inlet profile $u = U_0$ and a cubic polynomial exit profile $u = U_0 \frac{(3\eta - \eta^3)}{2}$ where $\eta = \frac{y}{\delta}$.

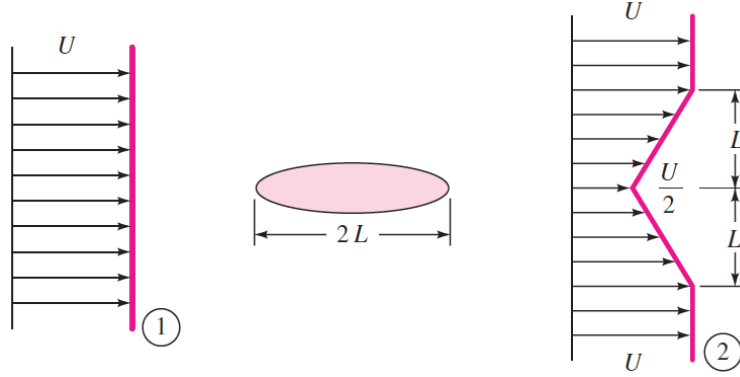
- Determine the volume flow Q across the top surface of the control volume.
- Determine the drag force exerted on the flow by the solid plate. Assume pressure is uniform, and hence there's no net pressure force.



Problem 2: Derive a formula for the horizontal force F required to hold the gate. At sections 1 and 2, the flow is uniform. Neglecting bottom friction. Assume steady incompressible flow with no variation across the width b . Express your final formula in terms of the inlet velocity V_1 , eliminating V_2 .



Problem 3: When a uniform stream flows past an immersed thick cylinder, a broad low-velocity wake is created downstream, idealized as a “V”. Pressures at 1 and 2 are approximately equal. If the flow is two-dimensional and incompressible, with width b into the paper, determine the drag force F exerted on the flow by the cylinder.



Problem 4: Consider water waves in a channel of depth H . Suppose that the span-wise width of the channel is large and the waves are two-dimensional. The undisturbed water level $y = 0$, is disturbed by the wave to $y = h(x, t) = A \cos(kx - \omega t)$, where A denotes the amplitude of the waves. The bottom of the channel is located at $y = -H$, and is rigid.

Suppose the velocity field due to the wave is given by:

$$\begin{aligned} u(x, y, t) &= -A\omega \sin(kx - \omega t) \cosh(k(y + H)) / \sinh(kH) \\ v(x, y, t) &= A\omega \cos(kx - \omega t) \sinh(k(y + H)) / \sinh(kH) \end{aligned}$$

1. Is this an incompressible flow?
2. Compute the vorticity field associated with this flow.
3. Find the pathlines for this flow. For this purpose take $A/H = 10^{-3}$ and $kH = 10$ and integrate the equation numerically (e.g. using MATLAB) and show the result as a graph of the computed pathlines (computer generated plots required). You can show the pathline originating at $x_0 = 0.5$, and $y_0 = 0.0$ at $t = 0$. Choose suitable non-dimensional variables in your calculation and plot, please define the choices you make. By doing this you should not need to know the values of the wavenumber k and frequency ω to plot the result.